

Systems and Methods for Big and Unstructured Data - Notes - v0.1.0

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Preface

Every theory section in these notes has been taken from the sources:

- Course slides. [1]

About:

 [GitHub repository](#)



These notes are an unofficial resource and shouldn't replace the course material or any other book on systems and methods for big and unstructured data. It is not made for commercial purposes. I've made the following notes to help me improve my knowledge and maybe it can be helpful for everyone.

As I have highlighted, a student should choose the teacher's material or a book on the topic. These notes can only be a helpful material.

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1 Big Data

1.1 The Big Data and AI Market Landscape

The first idea to establish is that **Big Data is not a single technology**. When people say “*Big Data*”, they may be referring to databases, distributed storage, analytics platforms, cloud infrastructure, machine learning, visualization tools, or complete applications. In reality, these components form an **ecosystem**. A useful simplified picture is:

Data Sources → Data Infrastructure → Analytics → Applications

Different companies and technologies operate at different points of this pipeline.

≡ Data sources: where everything starts

Before we can have “Big Data”, something must generate data. Modern systems continuously produce information from many heterogeneous sources like web, smartphones, IoT devices, social networks, transactions, sensors, and more.

Example 1: E-Commerce data sources

For example, imagine an e-commerce service. It might collect: customer profiles, purchases, searches, clicks, page views, product reviews images, payment transactions, delivery information. Notice immediately that **these are not all the same kind of data**. Some may be highly structured:

```
1 order_id = 58321
2 customer_id = 912
3 price = 49.99
```

While others may be much less structured:

“This product arrived late but works very well.”

Or could be images, video, GPS coordinates, logs, etc. This heterogeneity will later become one of the defining properties of Big Data: **Variety**.

≡ The infrastructure layer

Once data exists, we need technologies capable of **storing, transporting and processing it**. This is the infrastructure part of the Big Data market. Conceptually:

Raw Data → **Infrastructure** → usable data

Infrastructure contains things such as: distributed storage; databases; distributed computing systems; data ingestion systems; messaging systems; cloud infrastructure. Later in the notes we will encounter technologies belonging to this general world, such as distributed databases and large-scale processing architectures.

The fundamental challenge is that a **traditional architecture** such as *one database on one machine* may eventually become **insufficient**. Then the architecture can evolve toward something like:

Machine 1	Machine 2	Machine 3
data	data	data

But in the other hand, a **distributed architecture** introduces new questions about partitioning, replication, consistency, failures, and scalability. These questions will be central later in the notes.

🔗 Open-source platforms

Another important part of the Big Data ecosystem is the strong presence of **open-source platforms**. Historically, much of the Big Data revolution was enabled by systems that made distributed storage and computation available without requiring every company to implement these mechanisms from scratch. Instead of an organization implementing distributed file storage, fault recovery, parallel computation, cluster scheduling, data partitioning from zero, it can **build on an existing platform**.

So we can think of these platforms as **providing a reusable technological foundation**:

Distributed infrastructure → platform → data applications

A famous historical example that we will study later is the *Hadoop* ecosystem.

📊 Analytics layer

Simply storing huge quantities of data does not create value. Imagine a company has 100 TB of customer data. If nobody can derive useful information from it, those 100 TB are mostly a **cost**.

We therefore need another layer:

Stored Data → **Analytics** → Information / Models / Predictions

Analytics can include things such as: statistics; aggregation; data mining; machine learning; recommendation systems; forecasting; anomaly detection.

🔗 Example 2: Netflix analytics

Suppose Netflix records:

- User A watched movie X
- User A paused at 32 minutes
- User A finished movie Y
- User A abandoned movie Z

The **infrastructure problem** is: “*Can we reliably collect and store billions of these events?*”. The **analytics problem** is: “*Can we infer something useful from them?*”. For instance:

$$P(\text{user likes movie } m \mid \text{previous behaviour})$$

That distinction between **managing data** and **extracting information from data** is important.

Application layer

At the top, analytics becomes something the user or organization can actually use. For example:

behaviour data → recommendation model → “Recommended for you”

The user does not see the distributed storage system, cluster, database, or machine-learning infrastructure. They see the **application**. Other examples might include:

traffic data → traffic prediction → route recommendation

Or:

financial transactions → fraud detection model → fraud alert

Or:

sensor readings → predictive model → maintenance warning.

So there is a **transformation**:

raw data → processed data → insight → action

Full data-stack providers

Not every company operates at only one layer. Some vendors provide an almost complete data stack. Instead of offering only a database, or only analytics, they might offer:

storage + processing + analytics + ML + visualization + application services

These are the **full data stack providers**.

Example 3

Cloud platforms are an intuitive modern example of this idea.

A single provider may offer services for: object storage, databases, stream processing, distributed computing, data warehouses, ML training, model deployment, and business intelligence.

Therefore, the borders between the individual categories of the market are not strict.

🌱 Where does AI fit?

The AI market is strongly connected to this Big Data ecosystem because modern AI depends heavily on data infrastructure. A simplified relationship is:

Data → Data Engineering → Machine Learning / AI → Application

For example, suppose we want to train a recommendation model. The ML algorithm itself might be only one piece. Before training it, somebody must:

1. Collect user interactions;
2. Clean them;
3. Integrate different sources;
4. Store them;
5. Transform them into suitable features;
6. Feed them efficiently to the model.

Thus **AI requires a data system underneath it**. A powerful model with poor data infrastructure is often practically useless.

🔗 **Big Data and AI are therefore not synonyms.** This distinction is useful.

- **Big Data** is primarily concerned with questions such as: “*How do we collect, store, integrate, manage and process huge or complex datasets?*”
- **AI/ML** is more concerned with: “*How do we learn patterns or decision functions from data?*”

There is obviously a **large overlap** because **AI systems often consume enormous quantities of data**. But they solve different categories of problems. For Systems and Methods for Big and Unstructured Data (SMBUD), our main perspective is the **data-management side**. For example, before asking “*which neural network should we use?*”, we are much more likely to ask:

- *Where does the data live?*
- *How is it represented?*
- *How is it partitioned?*
- *How do we query it?*
- *Can the system scale?*
- *What consistency guarantees do we have?*

📌 Example 4: Spotify recommendation pipeline

Imagine Spotify wants to recommend songs. At the bottom we have **data sources**: play events, skips, likes, searches, playlists, artists, songs,

devices, locations. These might generate billions of events. Then:

1. **Data Sources** (user events, song metadata, etc.) generate data.
2. **Infrastructure** collects and stores them: Distributed Storage, Databases, Streaming Systems.
3. **Analytics layer** looks for patterns: $P(\text{user } u \text{ likes song } s)$.
4. **Application** shows: Discover Weekly, Recommended songs, Similar artists.

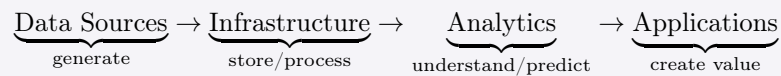
Therefore:

User events $\rightarrow$ Big Data infrastructure $\rightarrow$ Analytics/AI $\rightarrow$ Product feat

This is essentially the market landscape represented as a technical pipeline.

✓ Key Takeaways 1

We don't need to memorize the Big Data market as a collection of company logos. What matters is recognizing the **roles** inside the ecosystem:



With **open-source platforms** providing many of the underlying technological building blocks and **full-stack providers** potentially spanning several or all of these layers.

And the AI market sits naturally on top of and alongside this ecosystem because **without a scalable data pipeline, scalable AI is extremely difficult.**

References

- [1] Brambilla Marco. Systems and methods for big and unstructured data. Slides from the HPC-E master's degree course on Politecnico di Milano, 2026.